

Paper Reviewer Models

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TL;DR

Generate paper reviews using popular LLMs, and compare them with the official paper reviews from human experts. The primary goal is to identify any **inherent biases** present in the LLMs.

Ye, Rui et al.'s *exemplar paper* includes experiments that deal with paper structure to explore inherent flaws in LLMs

PDF: [Ye, Rui, et al. "Are we there yet? Revealing the risks of utilizing large language models in scholarly peer review." *arXiv preprint arXiv:2412.01708*, 2024.](#)

Section 2 Results, pages 8-9 are about the paper structure, which we also explore in our study. The results below show that LLMs favor lengthier papers and are prone to hallucinations.

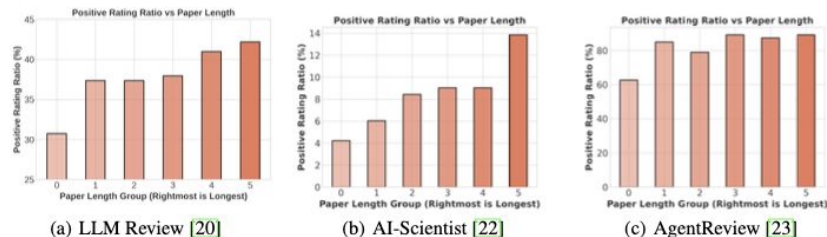


Figure 10: Examination of LLMs' preference of length in paper reviewing. Generally, all these review systems have preference towards longer papers.

Table 4: Averaged ratings of three review systems with varying input content. Papers with incomplete content (title, abstract, and introduction) could receive ratings that are comparable to papers with complete content, indicating an evident drawback of LLMs for serving as reviewers.

Content	LLM Review [20]	AI Scientist [22]	AgentReview [23]
Title Only	4.35 ± 0.98	2.96 ± 0.28	4.83 ± 0.60
Title + Abs. + Intro.	5.11 ± 0.58	3.36 ± 0.80	5.76 ± 0.42
Whole Paper	5.34 ± 0.60	3.41 ± 0.91	5.82 ± 0.38

Research Questions

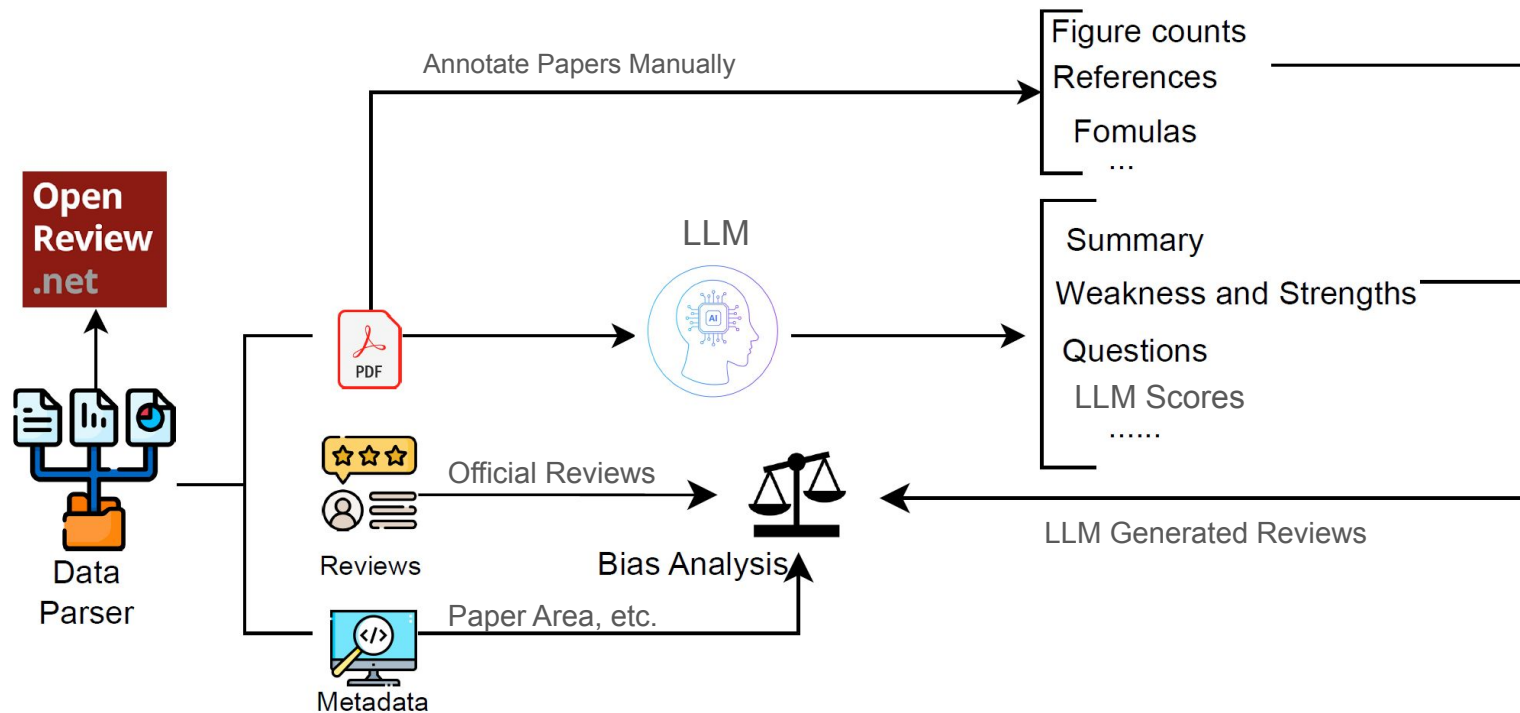
1. Are LLMs like ChatGPT and Gemini **reliable tools** for reviewing academic papers?
 - a. Is the difference between the scores assigned by the LLMs **significantly different** from those assigned by human expert reviewers?
2. Can we detect **inherent biases** in these LLMs with regards to the paper structure?
 - a. eg. number of figures, use of equations, amount of supplementary material, etc.
3. How can a UI **assist human reviewers** while not introducing any bias into the review decision?

Hypotheses

Our hypothesis is that there will be **some inherent bias** in the LLMs with regards to the paper structure, and that there will be some correlation between specific aspects of the paper structure and specific scores assigned.

Furthermore, we believe that we can propose a UI design that can **effectively assist human reviewers** in the review process.

Experiment Design



UI Design

Figma to mock the UI:

- **AI-Powered:** ChatGPT & Gemini intelligent review tools
- Split-view layout with original content and AI suggestions
- Real-time error highlighting and improvement tips
- One-click implementation of AI recommendations

NeurIPS 2024 Data was first sourced from **OpenReview.net**,
and the reviews were generated using LLMs

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1. Data was *obtained* from 2 categories: “Accept - Oral” and “Reject”
 - a. N = 61 papers from Accept Oral (the full set)
 - b. N = 21 papers from Reject (a subset)
 - c. We obtained the *paper pdfs* and the *official reviews* associated with each paper (Note: 4 official reviews per paper)
2. Data was *generated* using the *paper pdfs* and a custom prompt
 - a. 2 LLMs were used: *ChatGPT 4.0/4o* and *Gemini Flash 2.0*

Apparatus: ChatGPT 4.0/4o and Gemini 2.0 Flash were used to generate reviews using a standard *custom prompt*:

1. Task

You are an expert in reviewing papers submitted to the AI conference, NeurIPS. Please review the paper from the following aspects:

1. Summary

Briefly summarize the paper and its contributions. Focus on the motivation, key ideas, and achievements of the work. This summary should be neutral and factual, allowing authors to confirm that their work has been understood correctly. Avoid critique in this section.

2. Strengths and Weaknesses

Provide a thorough assessment of the paper's strengths and weaknesses, addressing the following dimensions:

- **Originality:** Are the tasks, methods, or ideas new? Is the work a novel combination of existing techniques? Does it clearly differentiate itself from prior work? Is related work adequately cited?
- **Quality:** Is the submission technically sound? Are claims well-supported by theoretical analysis, empirical results, or both? Are the methods appropriate? Is this a complete piece of work, or does it feel like work in progress? Are the authors honest about evaluating both strengths and weaknesses?
- **Clarity:** Is the submission clearly written and well-organized? Does it provide enough information for an expert reader to reproduce the results? If not, provide constructive suggestions for improving clarity.
- **Significance:** Are the results important? Are others likely to use or build on the ideas? Does the submission address a difficult task better than previous work? Does it advance the state of the art in a demonstrable way? Does it provide unique data, conclusions, or approaches?

3. Questions

List and carefully describe any questions or suggestions for the authors. Focus

on areas where a response from the authors could clarify confusion, address limitations, or change your opinion. This section is critical for a productive rebuttal and discussion phase.

4. Limitations

Evaluate whether the authors have adequately addressed the limitations and potential negative societal impact of their work. If not, provide constructive suggestions for improvement. Reward authors for being transparent about limitations and societal impact.

5. Ethical Concerns

Flag the paper for an ethics review if there are ethical issues. Refer to the NeurIPS ethics guidelines for guidance on when this is appropriate.

6. Soundness

Rate the soundness of the technical claims, experimental methodology, and research methodology on a scale of 1 to 4:

- 4: Excellent
- 3: Good
- 2: Fair
- 1: Poor

7. Presentation

Rate the quality of the presentation on a scale of 1 to 4, considering writing style, clarity, and contextualization relative to prior work:

- 4: Excellent
- 3: Good
- 2: Fair
- 1: Poor

8. Contribution

Rate the overall contribution of the paper to the research area on a scale of 1 to 4:

- 4: Excellent (important questions, significant originality, valuable results)
- 3: Good
- 2: Fair
- 1: Poor

9. Overall Score

Provide an overall score for the submission based on the following scale:

- 10: Award quality (groundbreaking impact, technically flawless, no ethical concerns)
- 9: Very Strong Accept (groundbreaking impact, technically flawless, excellent evaluation)
- 8: Strong Accept (novel ideas, excellent impact, strong evaluation)

- 7: Accept (technically solid, high impact, good evaluation)
- 6: Weak Accept (moderate-to-high impact, no major concerns)
- 5: Borderline Accept (reasons to accept outweigh reasons to reject)
- 4: Borderline Reject (reasons to reject outweigh reasons to accept)
- 3: Reject (technical flaws, weak evaluation, inadequate reproducibility)
- 2: Strong Reject (major technical flaws, poor evaluation, limited impact)
- 1: Very Strong Reject (trivial results, unaddressed ethical concerns)

10. Confidence Score

Rate your confidence in your assessment on a scale of 1 to 5:

- 5: Absolutely certain (very familiar with related work, checked details carefully)
- 4: Confident but not absolutely certain (unlikely but possible gaps in understanding)
- 3: Fairly confident (possible gaps in understanding, details not carefully checked)
- 2: Willing to defend assessment (likely gaps in understanding, details not checked)
- 1: Educated guess (submission not in your area or difficult to understand)

11. Code of Conduct Acknowledgement

Confirm that you agree to abide by the NeurIPS Code of Conduct while performing your duties as a reviewer.

Provide your review in the following JSON format:

```
{
  "Summary": "Brief summary of the paper and its contributions.",
  "Strengths and Weaknesses": {
    "Originality": "Assessment of the paper's originality.",
    "Quality": "Assessment of the paper's technical soundness and completeness.",
    "Clarity": "Assessment of the paper's clarity and suggestions for improvement.",
    "Significance": "Assessment of the paper's significance and potential impact."
  },
  "Questions": "List of questions and suggestions for the authors.",
  "Limitations": "Evaluation of how well the authors addressed limitations and societal impact.",
  "Ethical Concerns": "Flag for ethics review if applicable.",
  "Soundness": "Numerical rating for soundness (1-4).",
  "Presentation": "Numerical rating for presentation quality (1-4).",
  "Contribution": "Numerical rating for overall contribution (1-4).",
  "Overall Score": "Overall score for the submission (1-10).",
  "Confidence Score": "Confidence score for your assessment (1-5).",

  "Code of Conduct Acknowledgement": "Confirmation of agreement to abide by the NeurIPS Code of Conduct."
}
```

2. NeurIPS website instructions for official reviewers

3. Request for output to be in JSON 9

We manually annotated the data to check for different biases

Our Full Data:

Oral (Accept) papers: 61

- ChatGPT reviews: 61, Gemini Flash 2.0 reviews: 61, Official review: 61

Reject papers: 21

- ChatGPT reviews: 21, Gemini Flash 2.0 reviews: 21, Official review: 21

To condense the official reviews, we take the **average of the 4 review scores** for each paper.

Total Oral and Reject papers: 82

Total number of LLM reviews = 164 (including both ChatGPT and Gemini)

Total number of Official reviews = 82

Data Annotations:

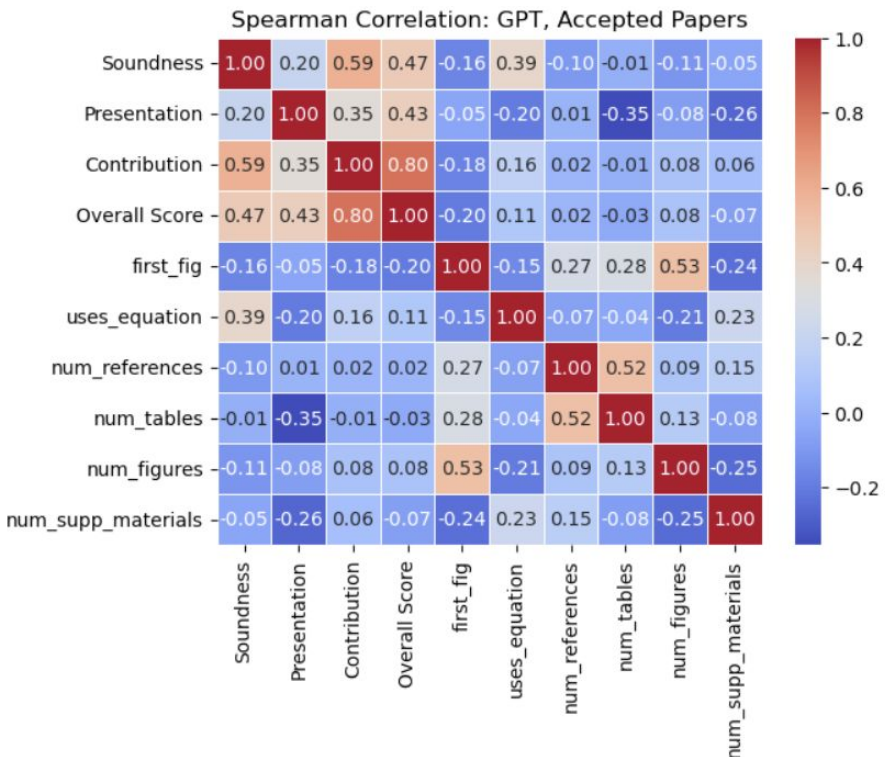
1. Figure on 1st or 2nd page
 - a. Binary value (i.e. 1 if paper has figure, 0 otherwise)
2. Number of figures
3. Number of references
4. Number of tables
5. Number of supplementary material pages
 - a. Count of pages after the references
6. Primary area (research)
 - a. The area of research the paper falls under (according to NeurIPS) obtained from website
7. Use of mathematical equations/formulas **as a proof**
 - a. *Binary value* (i.e. 1 if paper has equation as a proof, 0 otherwise)

Metrics, Measurements and Analysis

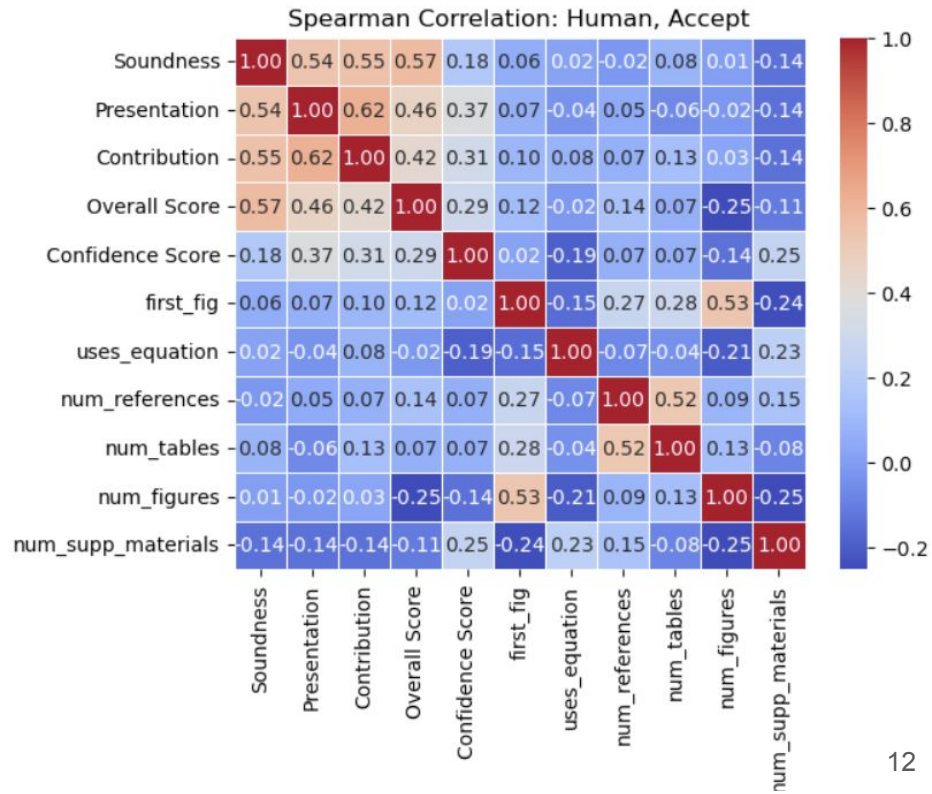
1. Visualize *positive or negative correlations* using the *Spearman Correlation Matrix*
 - a. Check correlations between the *model output scores* and the *annotated (bias) columns*
 - b. Compare the correlation matrices of the models with those of human expert reviews
2. The *Kruskal-Wallis test* is used to determine whether there is a statistically significant difference in Overall Scores across multiple independent groups (Primary Areas/Research Area)
 - a. Check for bias
3. Visualize the *distributions* of the model output scores and the official scores for oral-accept and reject, and check for statistically significant difference using either the *T-Test* or the *Mann Whitney U Test*

There is a positive correlation between the ChatGPT *soundness score* and the *use of equations*

ChatGPT



Human Reviewers

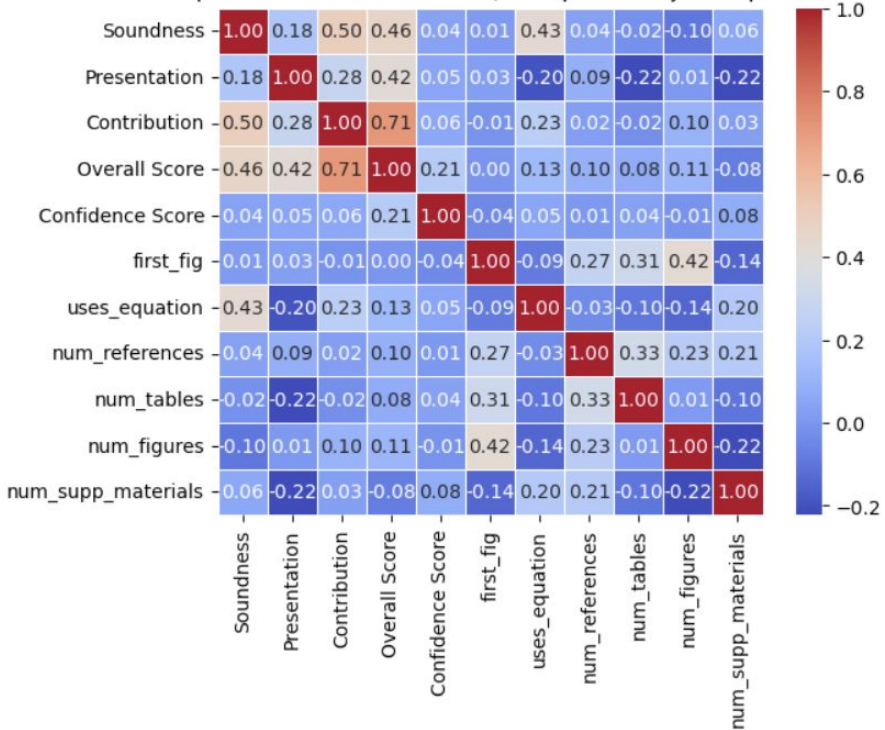


The positive correlation between the *soundness score* and the *use of equations strengthens* for the whole pool of papers (accept and reject)

ChatGPT

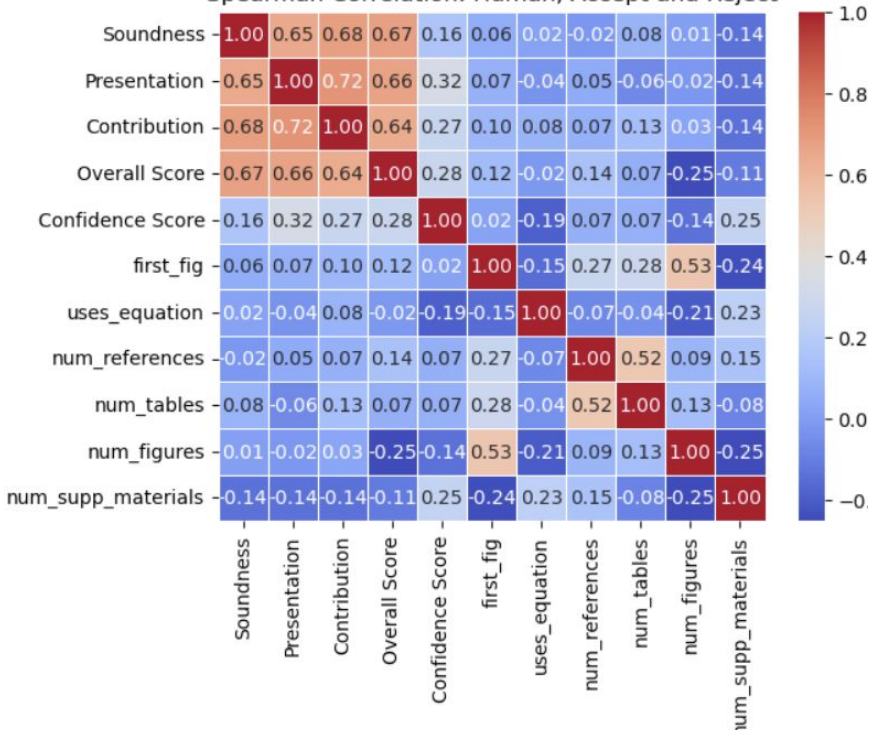


Spearman Correlation: GPT, Accept and Reject Papers



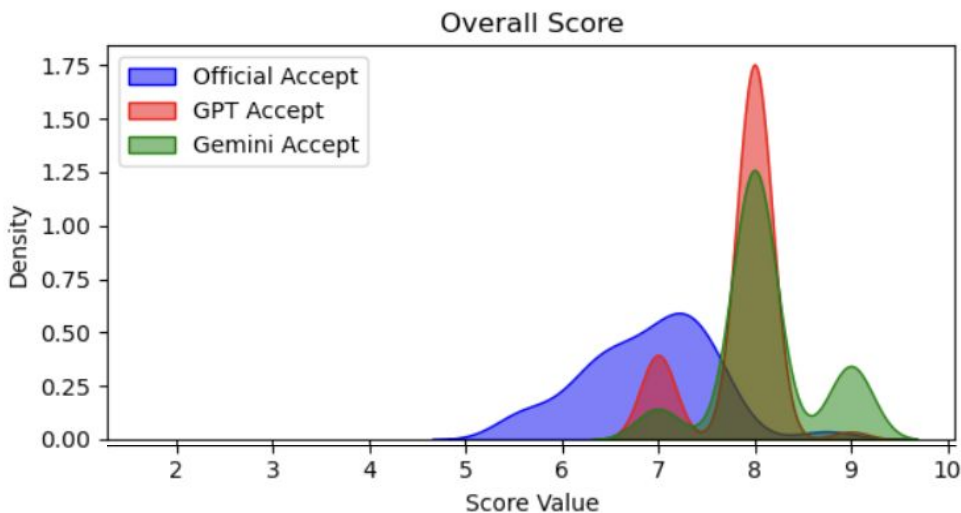
Human Reviewers

Spearman Correlation: Human, Accept and Reject



There is a significant difference between *all* the *HUMAN* review scores and *LLM* scores for *Accepted Papers*

Human vs GPT vs Gemini: Accept



ChatGPT vs Gemini showed significant differences between Overall Score, Presentation, and Confidence Score

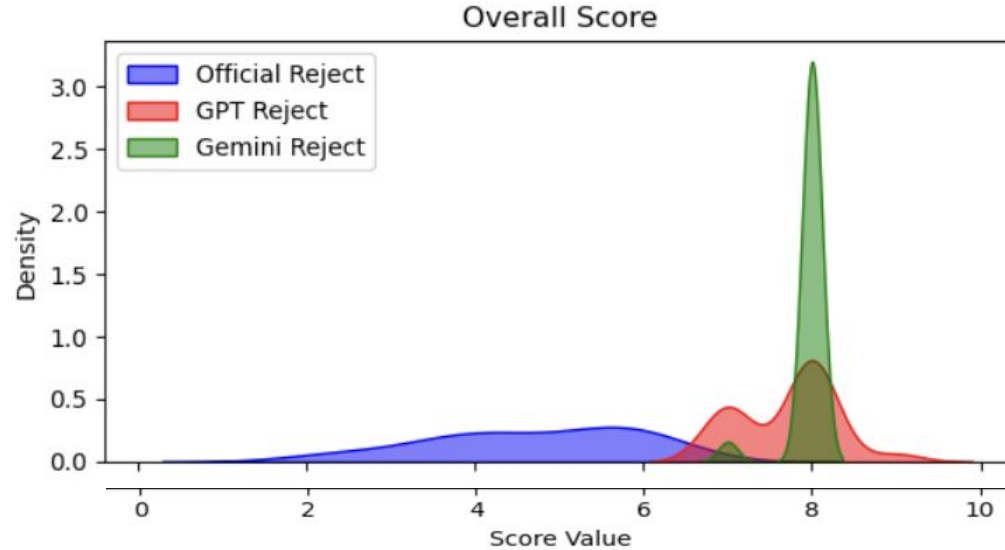
Human vs Gemini: ACCEPT			
Metric	P-Value	Significance	
0 Overall Score	2.41933e-19	Significant	
1 Soundness	2.11192e-19	Significant	
2 Contribution	1.10324e-18	Significant	
3 Presentation	2.83098e-16	Significant	
4 Confidence Score	1.70373e-18	Significant	

Human vs GPT: ACCEPT Papers			
Metric	P-Value	Significance	
0 Overall Score	9.49097e-16	Significant	
1 Soundness	1.8842e-20	Significant	
2 Contribution	7.56794e-18	Significant	
3 Presentation	0.0294917	Significant	
4 Confidence Score	0.000687984	Significant	

gpt vs gemini: Accept papers			
P-values from Mann-Whitney U Test:			
Metric	P-Value	Significance	
0 Overall Score	0.00176109	Significant	
1 Soundness	0.702949	Not Significant	
2 Contribution	0.787106	Not Significant	
3 Presentation	2.41474e-06	Significant	
4 Confidence Score	8.27629e-19	Significant	

There is a significant difference between *all* the *HUMAN* review scores and *LLM* scores for **Rejected Papers**

Human vs GPT vs Gemini: Reject



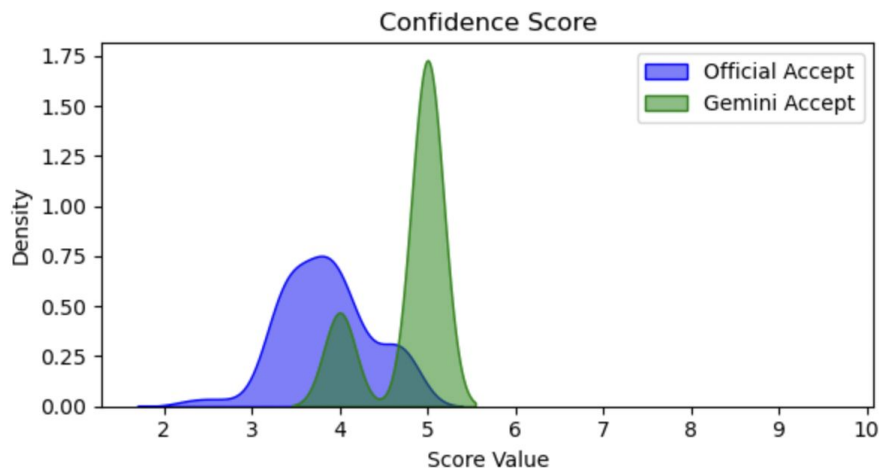
ChatGPT vs Gemini showed significant differences between Presentation and Confidence Score only

Human vs GPT: REJECT			
Metric	P-Value	Significance	
0 Overall Score	1.76094e-08	Significant	
1 Soundness	2.86621e-08	Significant	
2 Contribution	2.40541e-08	Significant	
3 Presentation	3.64457e-05	Significant	
4 Confidence Score	0.000770464	Significant	

Human vs Gemini: REJECT			
Metric	P-Value	Significance	
0 Overall Score	4.56126e-09	Significant	
1 Soundness	2.98981e-09	Significant	
2 Contribution	6.71399e-09	Significant	
3 Presentation	1.78357e-08	Significant	
4 Confidence Score	9.01876e-07	Significant	

gpt vs gemini: Reject papers			
P-values from Mann-Whitney U Test:			
Metric	P-Value	Significance	
0 Overall Score	0.0639689	Not Significant	
1 Soundness	0.162206	Not Significant	
2 Contribution	0.310586	Not Significant	
3 Presentation	4.25111e-08	Significant	
4 Confidence Score	5.74682e-05	Significant	

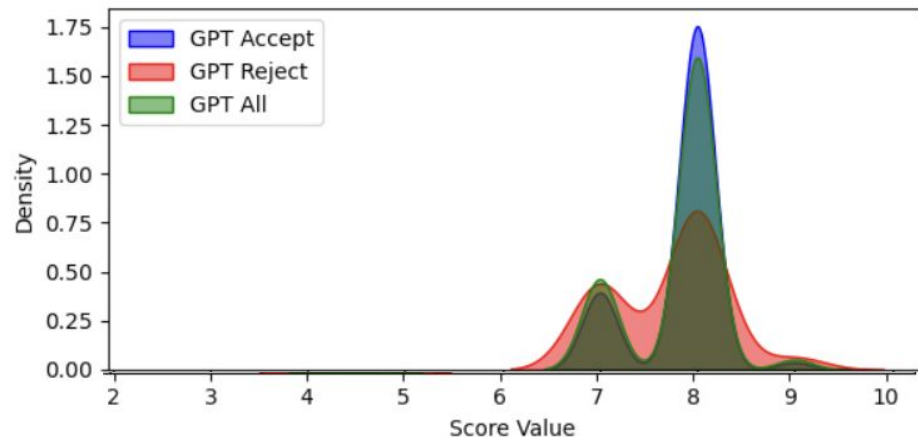
Interesting observation: Contrary to the results presented by *Liang et al.*, our findings show that LLMs typically **output higher confidence scores** than human reviewers



There is *no significant difference* between the **LLM overall scores for accepted paper** and **the LLM overall scores for rejected papers**

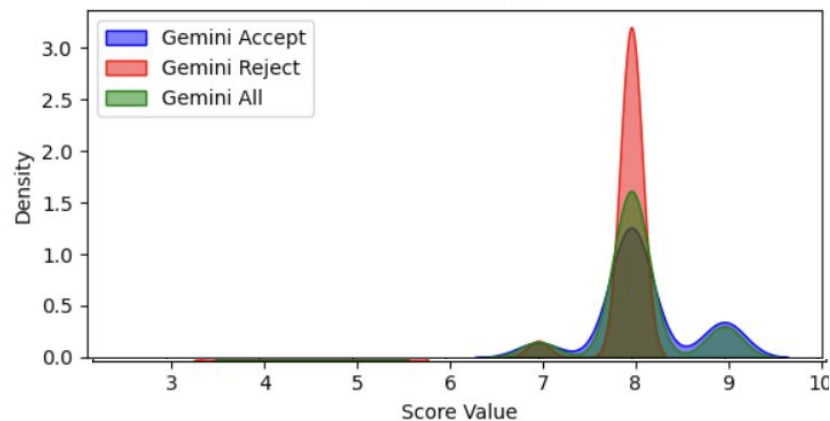
GPT Overall Score Distribution

Overall Score



Gemini Overall Score Distribution

Overall Score



-----gpt accept vs gpt reject-----

P-values from Mann-Whitney U Test:

	Metric	P-Value	Significance
0	Overall Score	0.25788	Not Significant
1	Soundness	0.457299	Not Significant
2	Contribution	0.899755	Not Significant
3	Presentation	0.00109173	Significant
4	Confidence Score	0.0161533	Significant

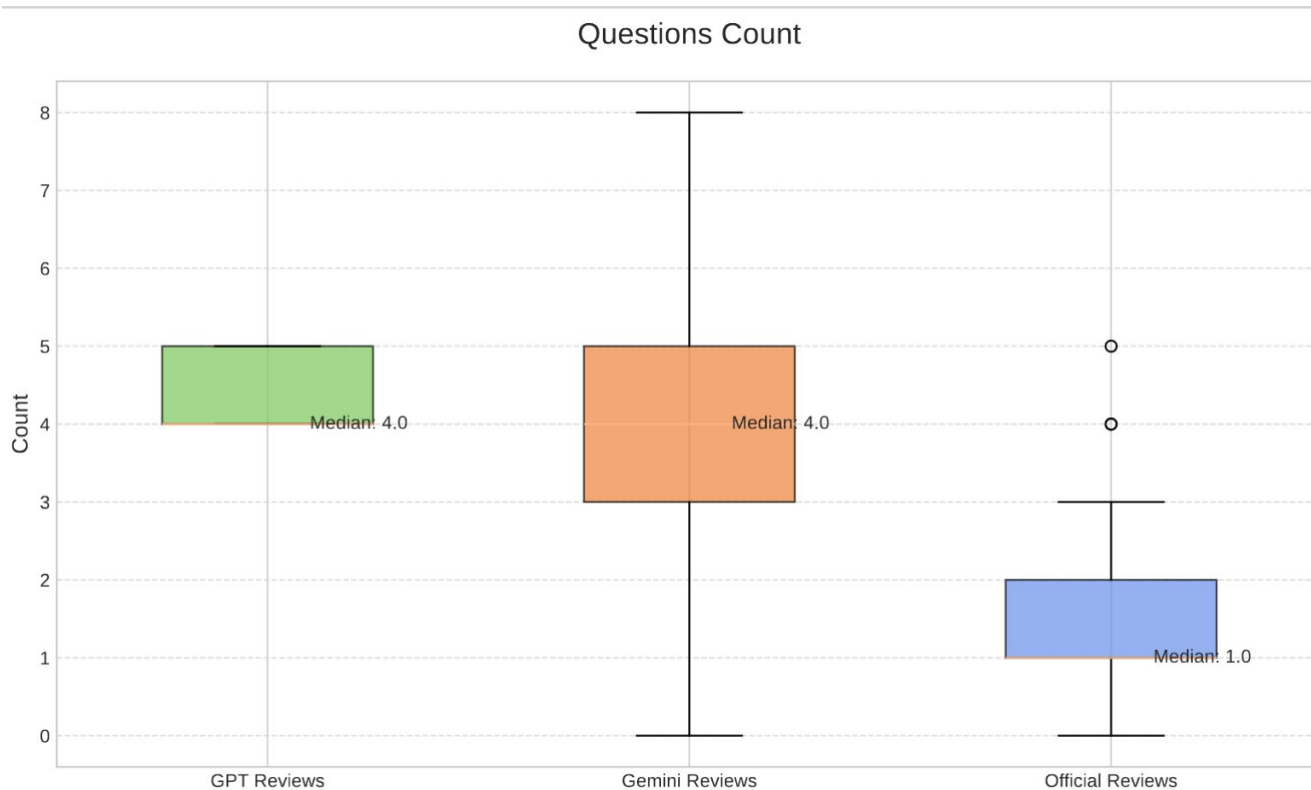
*For ChatGPT
there is a
significant
difference in
Presentation and
Confidence Score*

-----gemini accept vs gemini reject-----

P-values from Mann-Whitney U Test:

	Metric	P-Value	Significance
0	Overall Score	0.15132	Not Significant
1	Soundness	0.237366	Not Significant
2	Contribution	0.37969	Not Significant
3	Presentation	0.552604	Not Significant
4	Confidence Score	0.503709	Not Significant

LLMs are inclined to propose **more questions** than humans, and **limit its reviews** to the prompt constraint



AI-generated reviews and human written reviews have similarities, but still show **notable differences in the words usage**



AI-generated summary



LLMs write pros and cons according to **specified dimensions**, while humans **focus more on the details**

```
"Strengths and Weaknesses": {  
  "Originality": "The paper presents a novel approach to diffusion model guidance, using a guidance version of the same model instead of a separately trained unconditional model. This approach integrates guidance mechanisms and disentangles image quality improvements from prompt conditioning, distinguishing it from prior approaches such as classifier-free guidance and guidance intervals cited.",  
  
  "Quality": "The paper provides a thorough theoretical motivation and empirical validation to demonstrate state-of-the-art performance in multiple experimental setups, including unconditional settings. The authors carefully ablate different components, validating the effectiveness of their proposed method. However, while the experiments are extensive, a deeper analysis of the conditions under which autoguidance works best would strengthen the paper.",  
  
  "Clarity": "The paper is clearly written and well-structured. The explanations of the proposed guidance, and autoguidance are detailed and well-supported with visualizations. While the example analysis, could be made more intuitive for non-expert readers. The paper's reproducibility.",  
  
  "Significance": "The proposed approach is highly significant for the field of diffusion models by improving diffusion model performance without sacrificing sample diversity. The experimental metrics, along with qualitative analysis, suggest that this method could have a significant impact on future research. The ability to apply autoguidance to unconditional models is particularly noteworthy in areas where diffusion models struggle."  
},
```

```
"strengths": {  
  "value": "1. The paper employs an intuitive toy model to support its empirical findings. The tree-shaped mixture of Gaussian distributions used in this model could be beneficial for understanding generative models.\n\n2. The empirical observations and proposed methods are well-detailed, providing a clear mechanism within CFG and enhances it through the proposed method.\n\n3. The paper's findings are powerful, significantly improving the SOTA generation quality on the ImageNet dataset, which has the potential to inspire further related research."  
},  
"weaknesses": {  
  "value": "1. For the experiment, the paper lacks some quantitative comparison with the baseline text-to-image diffusion model."  
},
```

Summary of Our Main Findings

1. The *positive correlation* between the 'soundness' score and 'uses equation' is indicative of an *inherent bias in ChatGPT*.
2. The *significant differences* between human reviews and both LLM model reviews indicate that *neither ChatGPT nor Gemini scores align closely with human scores*.
3. For *accepted (strong) papers*, ChatGPT and Gemini *differ significantly* in their overall review scores.
4. For *rejected (weak) papers*, ChatGPT and Gemini overall scores are *not significantly different*.
5. *No significant differences* were found between ChatGPT or Gemini Overall Scores and *Primary Area (area of research)* using the Kruskal-Wallis test
6. LLMs are *inclined to follow the constraints of prompt and propose 3-4 questions*, while humans' are not limited. Humans may also write "see in weakness above".

Summary of Our Main Findings

7. LLM-generated review and human written reviews have similarities, but still *show differences in the words usage*; also, *humans are more rigorous*.
8. LLM-generated reviews write pros and cons according to specified dimensions, while humans focus *more on the details without a fixed format*.
9. *No significant difference* in the LLM Overall Score between Accepted and Rejected papers, so *both categories of papers are being scored very similarly*

Our UI Design

